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8.2 Primary and Secondary Metabolites

Most plants, fungi and other microbes synthesizes a number of organic compounds. These components are called as **metabolites** which are intermediates and products of metabolism. The term metabolite is usually restricted to small molecules. It can be catergorized into two types namely primary and secondary metabolites based on their role in metabolic process (Figure 8.5).

Primary metabolites are those that are required for the basic metabolic processes like photosynthesis, respiration, protein and lipid metabolism of living organisms.

Secondary metabolites does not show any direct function in growth and development of organisms.

Metabolites	Examples
Primary	
Enzymes	Protease, lipase, peroxidase
Amino acid	Proline, leucine
Organic acid	Acetic acid, lactic acid
Vitamins	A, B, C
Secondary	
Pigments	Carotenoids, anthocyanins
Alkaloids	Morphine, codeine
Essential oil	Lemon grass oil, rose oil
Toxins	Abrin, ricin
Lectins	Concanavalin A
Drugs	Vinblastin, curcumin
Polymeric substances	Rubber, gums, cellulose



Morphine is the first alkaloid to be found. It comes from the plant Opium poppy (*Papaver somniferum*). It is used as a pain reliever in patients with severe pain levels and cough suppressant.



8.2.1 Organic Molecules

Organic molecules may be small and simple. These simple molecules assemble and form large and complex molecules called **macromolecules**. These include four main classes – carbohydrates, lipids, proteins and nucleic acids. All macromolecules except lipids are formed by the process of polymerisation, a process in which repeating subunits termed monomers are bound into chains of different lengths. These chains of monomers are called **polymers**.

8.3 Carbohydrates

Carbohydrates are organic compounds made of carbon and water. Thus one molecule of water combines with a carbon atom to form CH_2O and is repeated several (n) times to form $\mathbf{(CH_2O)_n}$ where n is an integer ranging from 3–7. These are also called as **saccharides**. The common term sugar refers to a simple carbohydrate such as a monosaccharide or disaccharide that tastes sweet are soluble in water (Figure 8.7).